

Vinay Premchandran Nair

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EDUCATION

Carnegie Mellon University (CMU)

Pittsburgh, U.S.

Master of Science in Artificial Intelligence and Innovation (MSAI)

Aug 2022 - May 2024

• Coding Bootcamp, Distributed Systems, AI Engineering, ML in Production, Deep Learning, Machine Learning, NLP, Multimodal ML

Pune Institute of Computer Technology (PICT)

Pune, India

Bachelor of Engineering in Information Technology

Aug 2018 - July 2022

• Data Structures & Algorithms, Object Oriented Programming, Database Management, Cloud Computing, Computer Architecture

RELEVANT WORK EXPERIENCE

Senior Data Scientist | Apple, Inc. U.S.A. (V) | Fractal.ai

June 2025 - Present

- Built a **LangGraph**-based multi-agent framework that cut **apple.com** analysts' turnaround from 2–3 days to **under 15 minutes**.
- Developed a scalable memory layer that indexed **millions** of Snowflake records and **hundreds** of Keynote decks for the **async** "experience" **planner** agent performing dual **OpenSearch** lookups (structured vs unstructured indexes) to **dynamically route** queries to the right analytical tool, improving routing accuracy by **~30 %** and ensuring context-aware results.
- Developed a secure Snowflake MCP supporting 50+ analysts and ~1 K async queries/day under Apple's auth controls.
- Migrated conversation memory to **Redis** (async client) for TTL-based recall and **~1.8×** faster multi-turn context retrieval.
- Implemented **full async orchestration** for all **LLM**, **OpenSearch**, and agent calls, leveraging event loops in LangGraph and **Splunk**-instrumented logs to cut latency **~45 %** and monitor system performance under concurrent load.

Machine Learning Engineer (Capstone Project) | Bank of New York Mellon, U.S.A

Jan 2024 - May 2024

- Developed BNY Mellon's **Knowledge Graph** based **RAG** using **Neo4J** and **QDrant**, achieving a **10%** improvement in **reasoned retrieval** with novel **Contriver** embeddings augmenting the graph over regular vector databases using **LlamaIndex**
- Enhanced Financial **Multihop QA** accuracy by **2%** through Embeddings Augmented Knowledge Graph
- Boosted accuracy by **20%** in automated financial responses through **Microsoft Autogen** for **multi-agent LLM interactions**

Data Scientist | C3.ai, U.S.A.

Jan 2025 - June 2025

- Spearheaded a hybrid **RAG** pipeline in **PostgreSQL** using the **PGVector** plugin, leveraging **GIN indexing** for keyword filtering, **B-Tree** for metadata filtering, and **IVFFlat** indexing for vector search, boosting retrieval precision by **40%**.
- Built a user-configurable reranker (cross-encoder & LLM-based), improving passage ranking by **25%** across filtered retrievals.
- Built a Root Cause Analysis tool for RAG failures, saving **45–50 mins/query** of debugging and reducing manual effort by **80%**.

Machine Learning Engineer | Capgemini, U.S.A.

July 2024 - Jan 2025

- Enhanced legal document metadata extraction by **30%** using **AWS Textract** and **Claude 3.5**, processing over **1000** documents monthly
- Streamlined metadata storage and retrieval with AWS Lambda and **DynamoDB**, reducing process time by **25%**
- Developed a **ReactJS** web application with **AWS Lambda** integration, supporting daily operations for over **100** users
- Enhanced system monitoring with AWS CloudWatch and Splunk, managing 3,000+ operational events daily for optimal performance

RELEVANT PROJECTS

Tip Of the Tongue (ToT) Retrieval using LLMs (Advisors: Dr. Chenyan Xiong and Dr. Daphne Ippolito)

- Designed a **GPT-4** based **Query Decomposition** framework that improved query performance for movie search engines
- Implemented a **Dense Retrieval Model** to increase search efficiency and accuracy in matching movie names
- Developed a **Pointwise Reranking Mechanism** with GPT-4, independently scoring retrieved results for optimized ranking
- Improved key metrics, raising NDCG from **10% to 23%** and Precision from **4% to 20%** in search results

Augmenting Multimodal Multihop Question Answering (Advisor: Dr. Louis-Philippe Morency)

- Implemented a patch-focused **Vision-Language Transformer** and **T5** answer generator, resulting in an **8%** QA accuracy gain
- Implemented a **hierarchical approach for dense data representation**, improving multimodal reasoning capabilities by **2%**
- Engineered **MiniGPT-4** with Vision Transformer and Llama-V2 for advanced multimodal **Chain of Thought Reasoning**